SFT V12.12A: Photon Closure-Fit Addendum

The 4π Compton-Wavelength Bridge

SFT Research Notes

2026-06-16T22:43:42Z

PASS_PHOTON_COMPTON_WAVELENGTH_EQUALS_FERMION_4PI_CLOSURE_PATH
SETTLEMENT_FORCE_REMAINS_OPEN

Abstract

This addendum strengthens the V12.12 particle program by identifying a precise geometric bridge between the open photon branch and the closed spin-half fermion branch. For any spin-half branch obeying $r_{\text{spin}} = \hbar/(2mc)$, a photon with rest-energy scale $E_\gamma = mc^2$ has wavelength $\lambda_\gamma = h/(mc)$. But $4\pi r_{\text{spin}} = h/(mc)$. Therefore $\lambda_\gamma(E = mc^2) = 4\pi r_{\text{spin}}$. The photon does not merely have enough energy; its open longitudinal phase scale matches the full 4π spinor closure path of the corresponding fermion branch. This solves the geometric fit, not the microscopic settlement force.

1 The identity

For a V12.12 spin-half branch,

$$r_{\text{spin}} = \frac{\hbar}{2mc}.$$

The ordinary spatial length of one 2π loop is

$$2\pi r_{\text{spin}} = \frac{h}{2mc} = \frac{\lambda_C}{2}.$$

Thus one ordinary turn is half the Compton wavelength.

The full spinor closure is 4π:

$$L_{4\pi} = 4\pi r_{\text{spin}} = 4\pi \frac{\hbar}{2mc} = \frac{h}{mc} = \lambda_C.$$

For a photon at rest-energy scale $E_\gamma = mc^2$,

$$\lambda_\gamma = \frac{hc}{E_\gamma} = \frac{h}{mc} = \lambda_C.$$

Therefore

$$\boxed{\lambda_\gamma(E = mc^2) = L_{4\pi} = 4\pi r_{\text{spin}}.}$$

2 Electron numerical closure

For the electron,

$$m_e c^2 = 0.51099895 \text{ MeV}.$$

The rest-energy photon wavelength is

$$\lambda_\gamma \approx 2.426310239e - 12 \text{ m} = 2426.310 \text{ fm}.$$

The electron spin radius is

$$r_{\text{spin,e}} = \frac{\hbar}{2m_e c} \approx 1.930796339e - 13 \text{ m} = 193.080 \text{ fm}.$$

The full spinor closure path is

$$4\pi r_{\text{spin,e}} \approx 2.426310237e - 12 \text{ m} = 2426.310 \text{ fm}.$$

Thus a 0.511 MeV photon wavelength equals the electron full 4π spinor closure path. The required path curvature is

$$\kappa_{\text{path}} = \frac{1}{r_{\text{spin,e}}} \approx 5.179210e + 12 \text{ m}^{-1}.$$

This is path curvature, not ordinary weak-field spacetime curvature.

3 Pair-production wording

Free electron-positron pair production requires total center-of-momentum energy

$$2m_e c^2 \approx 1.022 \text{ MeV}.$$

Therefore the conservation-safe statement is: two head-on photons at the electron rest-energy scale can supply two matched 4π closure lengths, one for the electron branch and one for the positron branch.

A single isolated photon cannot become one free electron in empty space. For two null inputs,

$$M^2 c^4 = 2E_1 E_2 (1 - \cos\theta).$$

For equal head-on photons with $E_1 = E_2 = mc^2$ and $\theta = \pi$, $M = 2m$.

4 Mechanical interpretation

In V12.12 language:

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  open photon branch:  real energy at c with longitudinal phase scale lambda
closed fermion branch:  same energy folded into a 4pi spinor maintenance path
                        closure fit:  lambda = L_4pi
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Formation is not only energy accounting. It is geometric fit: the open phase length matches the closed spinor maintenance length.

5 Open settlement force

The closure-fit identity explains geometric compatibility. It does not derive the microscopic settlement force that keeps the branch closed.

The naive Newtonian diagnostic gives

$$E_G \sim \frac{G m_e^2}{r_{\text{spin,e}}} \approx 2.868e - 58 \text{ J}.$$

The electron rest energy is

$$m_e c^2 \approx 8.187e - 14 \text{ J}.$$

The ratio is

$$\frac{E_G}{m_e c^2} \approx 3.504e - 45.$$

Ordinary weak-field gravity is therefore short by a factor of about

$$2.854e + 44.$$

No factor-of-two relativistic correction can fix this. The 4π identity solves the closure-fit geometry; the stabilizing settlement law remains the deepest open problem.

The R183 stability targets remain:

$$E_{\text{env}}(r_0) = 0, \qquad E'_{\text{env}}(r_0) = \frac{Jc}{r_0^2}, \qquad E''_{\text{env}}(r_0) > -\frac{2Jc}{r_0^3}.$$

6 General branch table

branch	rest energy MeV	photon wavelength at E equals mc2 m	spin radius hbar over 2mc m	4pi spinor c
electron	0.51099895	2.426310e-12	1.930796e-13	2.426310e-1
muon	105.65838	1.173444e-14	9.337972e-16	1.173444e-1
tau	1776.86	6.977713e-16	5.552688e-17	6.977713e-1

7 Claim boundary

topic	claim	status	boundary
Closure-fit identity	For any spin-half branch with r_spin=hbar/(2mc), a photon with E=mc^2 has lambda=h/(mc)=4pi r_spin.	DERIVED_IDENTITY	Kinematic/geometric identity only; not a full formation amplitude.
Electron threshold meaning	A 0.511 MeV photon has wavelength equal to the electron branch's full 4pi spinor closure path.	SIGNATURE_RESULT	Free pair production still needs total center-of-momentum energy 2m_e c^2 and conservation-compatible geometry.
Pair-production threshold	Two head-on photons of electron rest-energy scale can supply two matched closure paths, one for e- and one for e+.	CONSERVATION_SAFE	Single isolated photon cannot become one free massive particle in empty space.
2pi vs 4pi	One ordinary 2pi spatial loop is lambda_C/2; full spinor closure is 4pi and equals lambda_C.	PRECISION_LOCK	Do not call the ordinary circumference the full Compton wavelength.
Photon extent	Wavelength is the longitudinal phase/closure-fit scale.	REALITY_LOCKED_INTERPRETATION	Do not claim every photon wavepacket has physical length exactly one wavelength.
Settlement force	The identity explains geometric fit, not the stabilizing force.	OPEN	Microscopic settlement law remains the deep open problem.
Ordinary gravity	Naive Gm^2/r is ~45 orders too weak for electron closure.	NEGATIVE_RESULT	No factor-of-two ultrarelativistic correction can fix this.

8 Equation index

id	equation	meaning
spin__radius	r__spin = hbar/(2mc)	Fermion branch centerline radius from V12.12/R176.
photon__wavelength	lambda__gamma(E=mc^2)=hc/(mc^2)=h/(mc)	Rest-energy photon wavelength.
closure__fit	lambda__gamma(E=mc^2)=4pi r__spin	Signature closure-fit identity.
ordinary__loop	2pi r__spin=lambda__C/2	One ordinary spatial loop is half the Compton wavelength.
full__spinor__path	4pi r__spin=lambda__C	Full spinor closure path equals the Compton wavelength.
path__curvature	kappa__path=1/r__spin=2mc/hbar	Path curvature required for closed c-speed branch.
two__null__threshold	M^2 c^4=2E1E2(1-cos theta)	R183 formation guardrail for two null inputs.
head__on__equal	E1=E2=mc^2, theta=pi => M=2m	Two equal head-on photons produce total invariant mass 2m.
naive__gravity	E__G ~ Gm^2/r__spin	Ordinary weak-field self-energy diagnostic.

9 Addendum statement

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At the rest-energy scale of a spin-half branch,
the open photon wavelength equals the full 4pi closed spinor path.
The photon-to-particle threshold is therefore a closure-fit condition:
open phase length matches closed maintenance length.
This identity explains geometric fit, not the microscopic settlement force.
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